CS 289A Fall 2026

Grad Project Proposal

Matthew Bernard *October 9, 2026*

Honor Code. I certify that all solutions in this document are entirely my own and that I have not looked at anyone else's solution. I have given credit to all external sources I consulted.

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1. Background

Application Domain. Post-training of foundation models is the dominant paradigm for adapting pre-trained models to downstream tasks. Unlike pre-training, which requires massive labeled corpora, post-training uses smaller, targeted data to align model behavior. Recent surveys categorize post-training into fine-tuning, alignment, reasoning, efficiency, and integration/adaptation (Chu et al. 2025). **Unsupervised Post-Training (UPT)** is a subset in which no ground-truth labels are used—the model generates its own training signal (Zhu et al. 2023). A central risk in UPT is *recursive error propagation*: pseudo-labels that are confidently wrong can reinforce themselves across training rounds (Zhu et al. 2023).

Overarching Goal. We will study whether **self-consistency**—the degree to which a Vision-Language Model (VLM) produces consistent outputs across multiple stochastic samples—can serve as a reliable unsupervised post-training signal for Visual Question Answering (VQA). Concretely:

> *Can we improve a VLM's VQA accuracy by post-training it on its own high-consensus outputs, without any ground-truth answers?*

We follow **Path 1** (post-training a foundation model), focusing on what makes post-training work best: the choice of features (sampling strategy), data (which pseudo-labels to retain), and hyperparameters (temperature, threshold, number of rounds).

2. Data and Resources

Data. We will use the **ScienceQA** dataset for unsupervised prompt selection and evaluation. - **Features:** image + text question. - **Labels:** multiple-choice answers, used *only* for evaluation, never as a training signal. - **Samples:** approximately 12{,}000 questions. - **Access:** publicly available via HuggingFace Datasets.

We will additionally evaluate on a held-out subset of **VQAv2** to test generalization.

Model. We will use **SmolVLM-500M-Instruct** as the foundation model, with a fallback to **LLaVA-1.5-7B** if compute permits.

Compute. A single GPU (A100 or L4) is sufficient. LoRA-based parameter-efficient fine-tuning keeps memory within 24,GB; full fine-tuning is feasible for SmolVLM-500M.

Unsupervised Signal. For each question, we sample k outputs (e.g., $k = 8$) at temperature $T > 0$. We compute the **mode** (most frequent answer) and retain questions whose mode frequency exceeds a threshold τ. The model is then fine-tuned (LoRA or full) on these self-generated pseudo-labels.

3. Timeline

Week	Milestone	Contributions
Oct 12–18	Environment setup; baseline zero-shot VQA evaluation on ScienceQA test set.	MB: model setup; teammate: data pipeline
Oct 19–25	Implement self-consistency sampling ($k = 8$, $T = 0.7$); measure baseline consistency distribution.	MB: sampling code; teammate: baseline analysis
Oct 26–Nov 1	Implement LoRA fine-tuning on self-consensus pseudo-labels; run first post-training round.	MB: training loop; teammate: data curation
Nov 2–6	Evaluate post-trained model; compare to baseline. **Progress Report due Nov 6.**	Both: evaluation and writing
Nov 7–13	Ablations: vary k, τ, temperature, LoRA rank, training epochs.	MB: ablations; teammate: analysis
Nov 14–20	Error-propagation study: does a second post-training round help or hurt?	Both: analysis
Nov 21–24	**First Draft of Paper due Nov 24.**	Both: writing
Nov 25–Dec 3	Revise paper; prepare peer review.	MB: revisions; teammate: peer review
Dec 4	**Peer Review due Dec 4.**	—
Dec 5–11	Final revisions based on reviews; record video. **Final Report + Video due Dec 11.**	Both: finalize

Internal milestones. Baseline accuracy (Oct 18), consistency–accuracy correlation plot (Oct 25), first post-trained checkpoint (Nov 1), ablation grid complete (Nov 13), error-propagation plot (Nov 20).

4. Evaluation Plan

We will report:

1. **Zero-shot baseline** accuracy under greedy decoding and majority-vote decoding.
2. **Post-trained accuracy** on the held-out test set, compared to baseline.
3. **Pseudo-label quality** as a function of τ (consistency threshold)—a key diagnostic.
4. **Ablation grid** over k, τ, T, LoRA rank, and number of training epochs.
5. **Error-propagation curve**: accuracy vs. number of post-training rounds.

Tradeoffs and failure modes (e.g., class imbalance induced by pseudo-label filtering) will be discussed explicitly, and final design decisions justified through ablations.

5. References

Chu, Tianzhe, Yuexiang Zhai, Jihan Yang, et al. 2025. "SFT Memorizes, RL Generalizes: A Comparative Study of Foundation Model Post-Training." *Proceedings of the 42nd International Conference on Machine Learning*, Proceedings of machine learning research, vol. 267: 10818–38. https://proceedings.mlr.press/v267/chu25c.html.

Zhu, Banghua, Jiantao Jiao, and Michael I Jordan. 2023. "Principled Reinforcement Learning with Human Feedback from Pairwise or k-Wise Comparisons." *arXiv Preprint arXiv:2301.11270*.